(1) Karageshwaran Dhakshinamoorthy

HW #14

Stat 50

Section 4.7

(1) $T \sim Exp(0.45)$

(a) $\mu_T = 1/\lambda = 1/0.45 = 2.2222$

(b) $\sigma_T^2 = 1/\lambda^2 = 4.9383$

(c) $P(T>3) = e^{-0.45(3)} = 0.25-9240261$

$P(T>3) = 0.2592$

(d)

$P(T>d) = 0.50 \Rightarrow e^{-0.45(d)} = 0.5$

$d = \ln(0.5)/-0.45$

$d = 1.5403$

(2) $\lambda = 0.25$

(a) $\mu_x = 1/0.25 = 4$

(b) $\sigma_x = \sqrt{1/\lambda^2} = 4 = \sqrt{1/0.06}$

(c) $P(x<3) = 1 - e^{-0.25(3)} = 0.5276$

(d) $P(x>11) = e^{-0.25(11)} = 0.0639$

(e) $P(x<k) = 0.50 = 1 - e^{-0.25(k)}$

$e^{-0.25(k)} = 0.5 \Rightarrow k = \ln(0.5)/-0.25$

(2)

$\lambda = 2.7726$

(f) $P(X \leq m) = 0.75 = 1 - e^{-0.25m}$

$m = \frac{\ln(0.25)}{-0.25}$

$\boxed{m = 5.5452}$

(g) $P(X < n) = 0.99 = 1 - e^{-0.25n}$

$n = \ln(0.01) / -0.25$

$\boxed{n = 18.4207}$

(h) $\mu = 12m$

(a) $\lambda = 1/\mu = 0.08333$

(b) $P(x < m) = 0.5 = 1 - e^{-0.08333a}$

$a = \frac{\ln(.5)}{-0.08333} = 8.3178$

(c) $\sigma_x = \sqrt{1/\lambda^2} = \sqrt{1/(1/12)^2} = 12$

$\boxed{\sigma_x = 12}$

(d) $P(x < l) = 0.65 = 1 - e^{-0.08333l}$

$l = \ln(0.35) / -0.08333 = 12.5979$

$\boxed{l = 12.5979}$

(7)

(a) $x \sim \text{poisson}(\lambda)$

$\lambda = \left(\frac{1}{12}\right) * 50 = 4.16667$

$P(x=5) = \dfrac{e^{-4.16667}(4.16667)^5}{5!}$

$\boxed{P(x=5) = 0.1623}$

(b) $x \sim \text{poisson}(\lambda)$

$P(x>2) = 1 - P(x \leq 2) = 1 - P(x=1) - P(x=2) - P(x=0)$

$\lambda = 20/12$

$= 1 - \dfrac{e^{-20/12}(20/12)^2}{2} - \dfrac{e^{-20/12}(20/12)^1}{1} - \dfrac{e^{-20/12}(20/12)^0}{0!}$

$= 0.7376727274 - 0.5096828274$

$= 0.2340$

(c) $\lambda = \left(\frac{1}{12}\right) * 15$

$P(x=0) = \dfrac{e^{-15/12}(15/12)^0}{0!}$

$= 0.2865$

(d) $P(x>2)$

(4)

(e) $P(8 < X < 20) = P(X < 20) - P(X \leq 8)$

(11)

(a) $\mu_T = 1/3$

(b) $\sigma_T = \sqrt{1/\lambda^2} = 1/\lambda = 1/3$

(c) $P(X > 1)$ $\quad X \sim Exp(3)$

$P(X > 1) = e^{-3(1)} = e^{-3} = 0.04979$

(d) $P(X < 1/12) = 1 - (P \geq 1/12) = 1 - e^{-3(1/12)} = 1 - e^{-(1/4)}$

$\boxed{P(X < 1/2) = 0.221199217}$

(e) $P(X < 1) = 1 - e^{-3(1)} = 0.9502$

(12)

(a) $\mu = 1/3$

(b) $P(X = 2) = \dfrac{(5/3)^2 e^{-5/3}}{2} = 0.2623$

(5)

(14)